

Lecture 6

Rotating elastic blade equations

ASDA/Dynamics of Rotor Systems

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QB 2.64 UoB

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This lecture

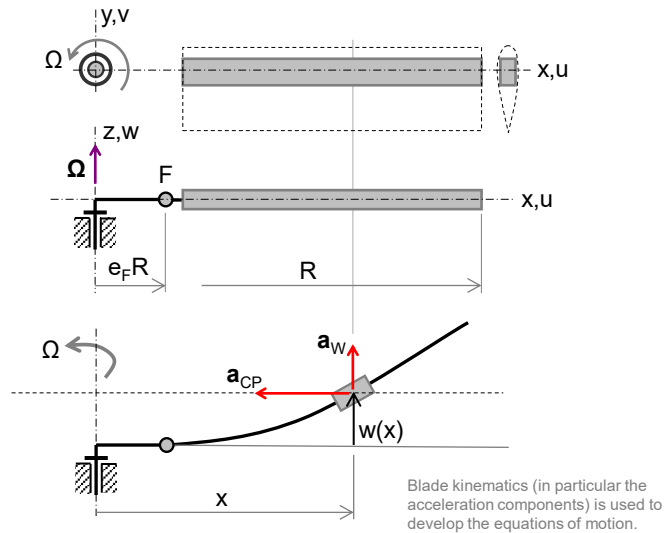
- Blade flapping equation
- Blade lead-lag equation
- Introduction to BVP
- Tension force distribution

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Kinematics of flapping motion

Basic notation used to describe elastic blade flapping:



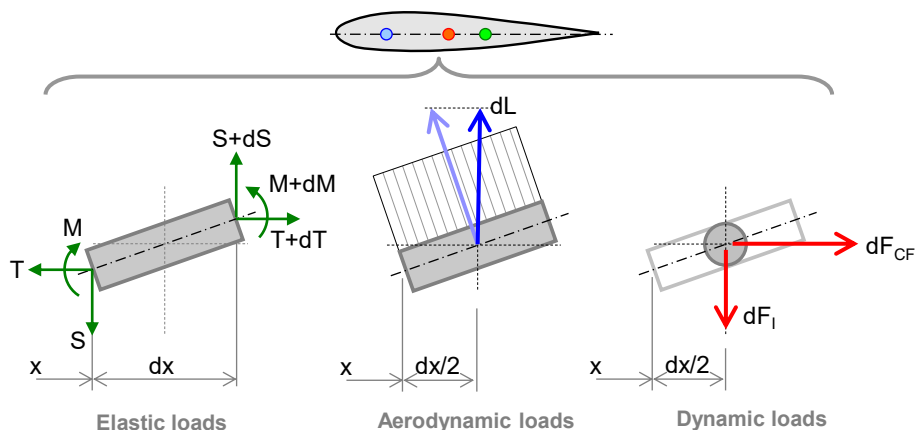
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Blade loads

We assume small deflections and pure flapping. Loads acting on a blade element:

- (i) elastic (internal) loads due to deformation (transversal shear S , longitudinal axial force T , bending moment M)
- (ii) aerodynamic (external) loads due to air flow (lift force L),
- (iii) dynamic (inertial) loads due to accelerated motion (inertial force F_i , centrifugal force F_{CF}).

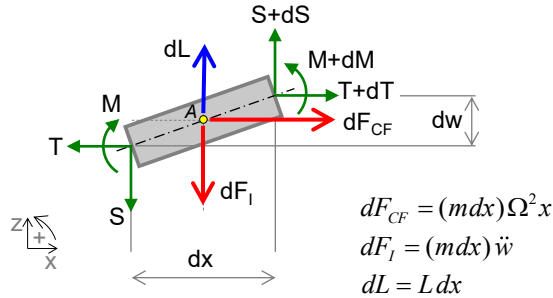


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Blade element dynamic equilibrium equations

Blade element with all loads superposed:



Dynamic equilibrium equations:

$$\begin{aligned} \sum_{(i)} F_{x,i} &= 0 & -S + S + dS + L dx - (mdx)\ddot{w} &= 0 \\ \sum_{(j)} F_{z,j} &= 0 & -T + T + dT + (mdx)\Omega^2 x &= 0 \\ \sum_{(k)} M_{A,k} &= 0 & -M + M + dM + S \frac{dx}{2} + (S + dS) \frac{dx}{2} - T \frac{dw}{2} - (T + dT) \frac{dw}{2} &= 0 \end{aligned}$$

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Blade element equations, continued ...

Force equilibrium:

$$\begin{aligned} dS &= (m\ddot{w} - L) dx & \Rightarrow \frac{dS}{dx} &= m\ddot{w} - L \\ dT &= -(mdx)\Omega^2 x & \Rightarrow T(x) &= \Omega^2 \int_x^R m(\tilde{x}) \tilde{x} d\tilde{x} \end{aligned}$$

Auxiliary equations arising from this derivation.

Moment equilibrium:

$$\begin{aligned} -M + M + dM + S \frac{dx}{2} + (S + dS) \frac{dx}{2} - T \frac{dw}{2} - (T + dT) \frac{dw}{2} &= 0 \\ dM + S dx + dS \frac{dx}{2} - T dw - dT \frac{dw}{2} &= 0 \\ \frac{dM}{dx} + S - T \frac{dw}{dx} &= 0 \end{aligned}$$

We neglect higher order terms (characterized by the products of already small quantities).

$$\frac{d^2 M}{dx^2} + \frac{dS}{dx} - \frac{d}{dx} \left(T \frac{dw}{dx} \right) = 0$$

The main outcome (equation) arising from this derivation.

$$M = EI \frac{\partial^2 w}{\partial x^2}$$

Apply the E-B constitutive law.

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Equation of blade flapping

The equation derived above describes flapping of the rotating elastic blade. This equation is based on the Euler-Bernoulli beam bending assumptions:

$$\frac{\partial^2}{\partial x^2}\left(EI_F\frac{\partial^2 w}{\partial x^2}\right)-\frac{\partial}{\partial x}\left(T\frac{\partial w}{\partial x}\right)+m\ddot{w}=L$$

Elastic term

Inertial term due to CF

Inertial term due to flapping

Aero term

This is a non-homogeneous partial differential equation (PDE) of the fourth order in x , with the time invariant coefficients and non-zero RHS. This is the equation of motion of rotating elastically flapping blade.

Other auxiliary and useful relationships (finding internal loads, defining boundary conditions, etc.):

Internal axial force: $T(x)=\Omega^2\int_x^Rm(\tilde{x})\tilde{x}d\tilde{x}$

Internal bending moment: $M=EI\frac{\partial^2 w}{\partial x^2}$

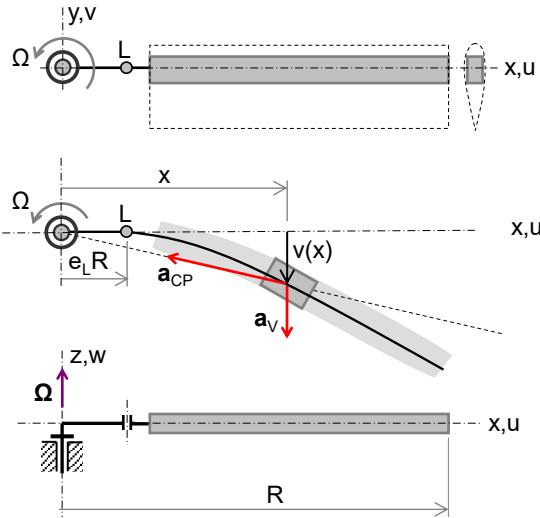
Internal shear force: $S=T\frac{dw}{dx}-\frac{dM}{dx}$

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Kinematics of lead-lag motion

Basic notation used for the elastic blade lead lag motion:

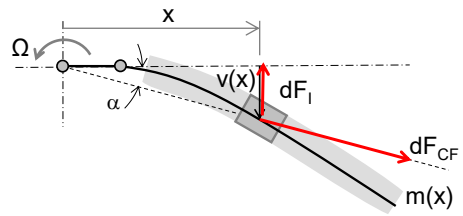


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Centrifugal stiffening

For the derivation of the lead lag EoM we use the EoM of elastic flapping with the modified term which represents the transversal loads.



From geometry, the CF load has a non-zero component in the transversal direction for this type of blade motion.

Instead of deriving the equation of motion for this scenario "from scratch", we strive to reuse the results of the previous derivation. To do that, first, we observe similarity and differences, in terms of the inertial loading, between the flapping and lead-lag cases.

Note that, in contrast with flapping, the CF load here manifests its influence not only in the y-direction but also in the perpendicular direction. This is a new influence compared to the flapping case, and it is further accounted for in the simplified derivation.

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Equation of blade lead-lag

Consider the inertial components of the transversal loads:

$$(D(x) - m\ddot{v})dx + m\Omega^2 v dx =$$

$$(D(x) - m(\ddot{v} - \Omega^2 v))dx$$

EoM for the elastic lead lag motion is derived based on the analogy with already derived EoM for elastic flapping:

$$\frac{\partial^2}{\partial x^2} \left(EI_F \frac{\partial^2 w}{\partial x^2} \right) - \frac{\partial}{\partial x} \left(T \frac{\partial w}{\partial x} \right) + m\ddot{w} = L$$

Euler-Bernoulli equation for the flapping motion of a rotating elastic blade

$$\frac{\partial^2}{\partial x^2} \left(EI_L \frac{\partial^2 v}{\partial x^2} \right) - \frac{\partial}{\partial x} \left(T \frac{\partial v}{\partial x} \right) + m(\ddot{v} - \Omega^2 v) = D$$

Euler-Bernoulli equation for the lead lag motion of a rotating elastic blade

Elastic term
Inertial term due to CF
Inertial term due to lead lag motion and Ω
Aero term

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The complete model of a rotating elastic blade

The uncoupled rotating elastic blade flapping and lead-lag model:

Equation of flapping motion:

$$(EI_F w'')'' - (T w')' + m \ddot{w} = L$$

Equation of lead-lagging motion:

$$(EI_L v'')'' - (T v')' + m(\ddot{v} - \Omega^2 v) = D$$

Blade tension distribution:

$$T = \Omega^2 \int_x^R m \tilde{x} d\tilde{x}$$

Partial Differential Equations (PDEs) of the fourth order in space and second order in time variables:

- four Boundary Conditions (BCs)
- two Initial Conditions (ICs)

A known function after the indicated integration completed. No unknown component present in this expression.

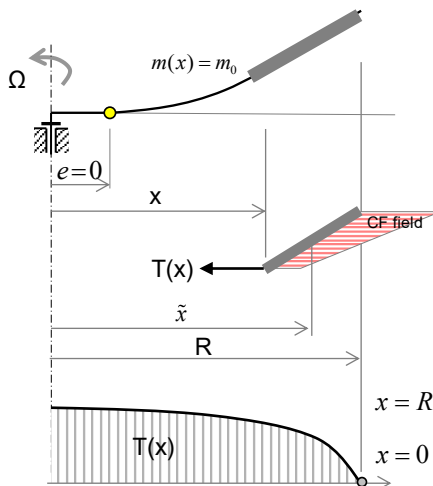
These are the key equations (PDEs) which will be used in the rest of the course. These equations need to be solved (*integrated*). The two dependent variables are the blade flapping response (w) and lead-lag response (v). The two independent variables are the time (t) and position along the blade (x). Mathematically, this is the **Initial Value Boundary Value Problem (IV-BVP)** and, to solve it, we need the **Initial Conditions** (initial blade deformation) and **Boundary Conditions** (blade conditions at its root and tip). After this, the solution can be used to determine the blade deformations and loads, natural frequencies, stability, aerodynamics loads, etc.

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Example: finding the blade root tension load

Determine the distribution of the tension force $T(x)$ due to rotation. The rotor speed is $\Omega=30$ rad/s, the rotor radius is $R=8$ m and the mass distribution is $m(x)=12$ kg/m. Determine the load in the flapping hinge.



$$T(x) = \Omega^2 \int_x^R m(\tilde{x}) \tilde{x} d\tilde{x}$$

$$T(x) = \Omega^2 m_0 \int_x^R \tilde{x} d\tilde{x} = \Omega^2 m_0 \left[\frac{\tilde{x}^2}{2} \right]_x^R$$

$$T(x) = \frac{1}{2} \Omega^2 m_0 (R^2 - x^2)$$

$$x = R : T(R) = 0$$

$$x = 0 : T(0) = \frac{1}{2} \Omega^2 m_0 R^2 = \frac{1}{2} 30^2 * 12 * 8^2 \approx 346 \text{ kN}$$

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Summary

- Dynamic equilibrium approach used to obtain the equations of motion further using:
 - Euler-Bernoulli assumptions.
 - Decoupled flap $w(x,t)$ and lead-lag $v(x,t)$.
 - Distributed tension from the rotation $T(x)$.
- EoMs = PDEs + additional conditions
 - Initial conditions ($t=0$) for the time component.
 - Boundary conditions ($x=\{\text{root}, \text{tip}\}$) for the span component.
- Resulting calculated deformations allow further computations:
 - Bending moments $M(x,t)$,
 - Shear forces $S(x,t)$.
- IV-BVP: solve PDE with two ICs and four BCs.

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